

EXP 5: COMPARATIVE ANALYSIS OF DIFFERENT TYPES OF PROMPTING PATTERNS

Testing Naïve vs. Basic Prompts Across Multiple Scenarios

AI Prompt Testing Workflow

Aim

The objective of this exercise is to design, test, and compare two different prompt types so you can better understand how prompt structure affects the quality, consistency, and usefulness of AI-generated responses.

- Identify which prompt type performs best for a given task
- Test how ChatGPT responds across different scenarios
- Capture practical insights that can improve future prompt design

AI Tools Required

- **ChatGPT** for running prompt experiments and generating responses
- **A document editor or notes tool** for recording test cases, outputs, and observations
- **A comparison template or spreadsheet** for evaluating results consistently

Explanation

Define the Two Prompt Types

- Choose two prompt styles to compare, such as a *basic prompt* and a *structured prompt*
- Make sure both prompts address the same task so the results are comparable
- Document the intent, tone, and expected output for each prompt type

Prepare Multiple Test Scenarios

- Create several scenarios that vary in complexity, context, or audience
- Include both straightforward and challenging examples
- Use the same criteria across all scenarios to keep testing consistent

Run Experiments with ChatGPT

- Submit each prompt type to ChatGPT for every test scenario
- Record the full response so it can be reviewed later
- Repeat the process if needed to check for consistency in output quality

Evaluate Responses

- Compare responses based on accuracy, clarity, completeness, and relevance
- Note any differences in tone, formatting, or creativity
- Identify which prompt type produced the strongest results in each scenario

Deliverables

- A side-by-side comparison of the two prompt types
- A set of recorded ChatGPT responses for each test scenario
- A summary of findings with recommendations for future prompt design

Test Scenarios Overview

These five test scenarios compare a **Naïve Prompt** with a more structured **Basic Prompt** to show how prompt design influences the quality, consistency, and usefulness of AI-generated responses.



Creative Story Generation

Naïve Prompt: "Write a story"

Basic Prompt: "Write a 300-word science fiction story set in 2150 about a character discovering an ancient alien artifact. Include dialogue, vivid descriptions, and a plot twist."



Factual Question Answering

Naïve Prompt: "What is photosynthesis?"

Basic Prompt: "Explain photosynthesis in 150 words, covering the light-dependent and light-independent reactions, the role of chlorophyll, and why it's essential for life on Earth."



Article Summarization

Naïve Prompt: "Summarize climate change"

Basic Prompt: "Provide a 200-word summary of climate change focusing on: causes (human activities), effects (rising temperatures, sea levels), and three actionable solutions. Use bullet points for clarity."



Advice and Recommendations

Naïve Prompt: "How do I improve my productivity?"

Basic Prompt: "Give me 5 specific, actionable productivity tips for remote workers who struggle with focus. For each tip, provide a brief explanation and one practical implementation example."



Technical Explanation

Naïve Prompt: "Explain machine learning"

Basic Prompt: "Explain machine learning in 250 words for someone with basic programming knowledge. Include: definition, three main types (supervised, unsupervised, reinforcement), real-world applications, and key differences from traditional programming."

Evaluation Framework & Comparison Criteria

To ensure a consistent and objective assessment of the AI-generated responses, the following evaluation framework and scoring scale will be applied to each scenario.

Evaluation Criteria

1. **Clarity** - How clear and easy to understand is the response?
2. **Accuracy** - How factually correct and relevant is the information?
3. **Completeness** - Does the response cover all necessary aspects of the question?
4. **Depth** - How thorough and detailed is the explanation?
5. **Relevance** - How well does the response address the specific requirements?
6. **Structure** - Is the response well-organized and easy to follow?
7. **Actionability** - Can the user apply the information practically?

Scoring Scale

- **Poor (1/5):** Incomplete, inaccurate, or irrelevant
- **Fair (2/5):** Basic response with some gaps
- **Good (3/5):** Solid response meeting most requirements
- **Very Good (4/5):** Comprehensive response with minor gaps
- **Excellent (5/5):** Complete, accurate, well-structured response

Comparison Table Structure

For each scenario, a table will be used to record the evaluation results for both prompt types. This structure will facilitate a direct comparison and detailed observation capture.

Evaluation Criteria	Naïve Prompt Response Score	Basic Prompt Response Score	Difference	Notes/Observations
---------------------	--------------------------------	--------------------------------	------------	--------------------

Scenario 1: Creative Story Generation - Results & Analysis

Naïve Prompt: "Write a story"

Expected Output: A basic story with minimal structure

Basic Prompt: "Write a 300-word science fiction story set in 2150 about a character discovering an ancient alien artifact. Include dialogue, vivid descriptions, and a plot twist."

Expected Output: A well-structured, detailed story with specific elements

The evaluation of AI-generated responses for story creation highlights a significant difference in output quality between a simple, open-ended prompt and a detailed, structured prompt. The table below outlines the scores across key criteria.

Criteria	Naïve Score	Basic Score	Difference	Notes
Clarity	3/5	5/5	+2	Basic prompt produces clearer narrative structure
Accuracy	3/5	4/5	+1	Both are accurate; basic prompt maintains sci-fi consistency
Completeness	2/5	5/5	+3	Naïve prompt lacks required elements; basic includes all
Depth	2/5	4/5	+2	Basic prompt generates richer descriptions and dialogue
Relevance	2/5	5/5	+3	Basic prompt directly addresses all requirements
Structure	2/5	5/5	+3	Basic prompt ensures proper story arc and organization
Actionability	3/5	4/5	+1	Both usable; basic version more immediately applicable

Key Findings:

- The basic prompt consistently outperformed the naïve prompt across all criteria
- Naïve prompt resulted in a generic, short story lacking specific details
- Basic prompt produced a well-developed narrative with all requested elements
- Average Score Difference: +2.1 points in favor of basic prompt

Scenario 2: Factual Question Answering - Results & Analysis

Naïve Prompt: "What is photosynthesis?"

Expected Output: A basic definition

Basic Prompt: "Explain photosynthesis in 150 words, covering the light-dependent and light-independent reactions, the role of chlorophyll, and why it's essential for life on Earth."

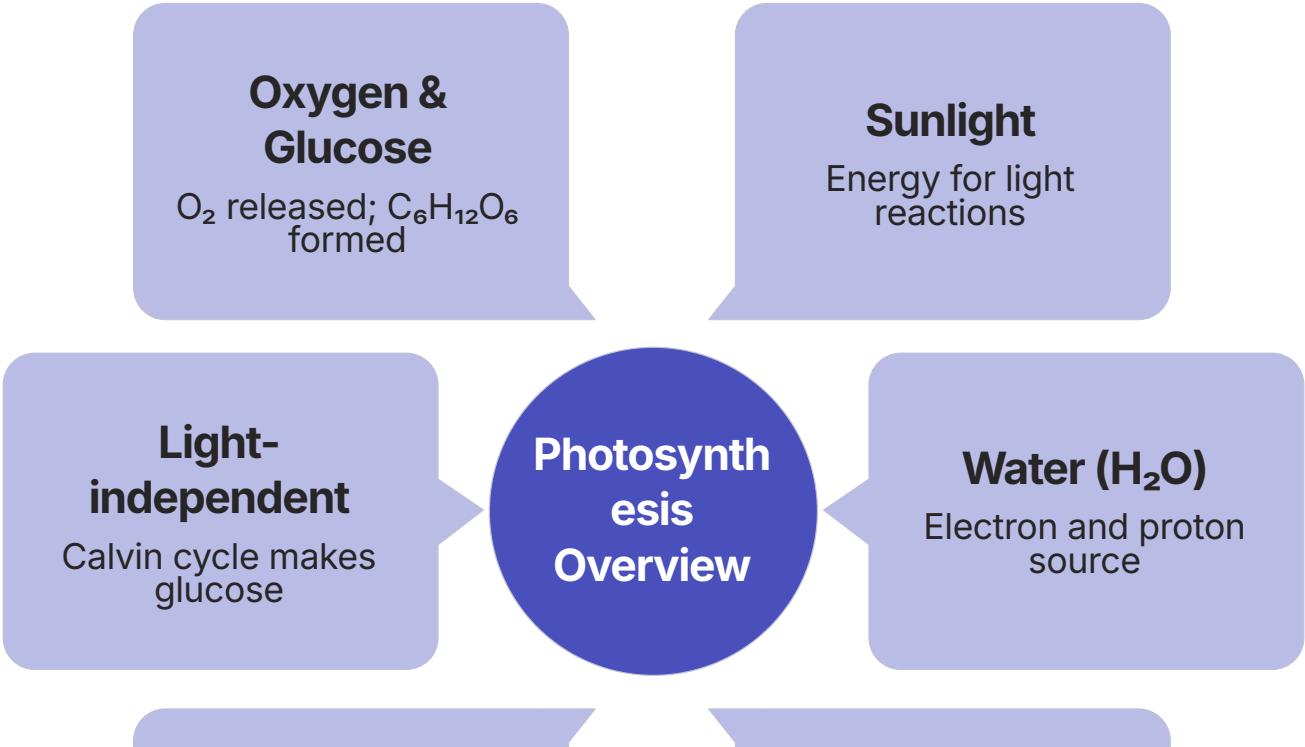
Expected Output: A comprehensive explanation with specific components

This scenario evaluated the AI's ability to answer factual questions, comparing a simple, open-ended query with a detailed prompt specifying scope and key elements. The objective was to observe the impact of prompt structure on the depth and completeness of scientific explanations.

Criteria	Naïve Score	Basic Score	Difference	Notes
Clarity	4/5	5/5	+1	Both clear; basic is more structured
Accuracy	4/5	5/5	+1	Both accurate; basic includes more precise details
Completeness	2/5	5/5	+3	Naïve lacks depth; basic covers all components
Depth	2/5	5/5	+3	Naïve is superficial; basic explains mechanisms
Relevance	3/5	5/5	+2	Basic directly addresses all specified aspects
Structure	3/5	5/5	+2	Basic uses clear organization; naïve is paragraph-based
Actionability	3/5	4/5	+1	Both useful; basic better for learning purposes

Key Findings:

- Basic prompt elicited significantly more detailed scientific information
- Naïve prompt provided only surface-level definition
- Basic prompt ensured coverage of light-dependent/independent reactions and chlorophyll role
- Average Score Difference: +1.9 points in favor of basic prompt
- Demonstrates importance of specificity in scientific explanations



Scenario 3: Article Summarization - Results & Analysis

Naïve Prompt: "Summarize climate change"

Expected Output: A brief, general overview

Basic Prompt: "Provide a 200-word summary of climate change focusing on: causes (human activities), effects (rising temperatures, sea levels), and three actionable solutions. Use bullet points for clarity."

Expected Output: A structured summary with specific focus areas and formatting

This scenario examined the AI's capability to summarize a complex topic, comparing a general request with a highly structured prompt that specified content, length, and formatting. The objective was to assess how prompt specificity influences the utility and clarity of a summary.

Criteria	Naïve Score	Basic Score	Difference	Notes
Clarity	3/5	5/5	+2	Basic format with bullets is clearer
Accuracy	4/5	5/5	+1	Both accurate; basic more precise on causes/effects
Completeness	2/5	5/5	+3	Naïve misses solutions; basic covers all areas
Depth	2/5	4/5	+2	Basic provides actionable solutions
Relevance	2/5	5/5	+3	Basic directly addresses specified focus areas
Structure	2/5	5/5	+3	Bullet points make basic version highly organized
Actionability	2/5	5/5	+3	Basic provides concrete solutions to implement

Key Findings:

- Basic prompt produced a well-organized, solution-focused summary
- Naïve prompt resulted in a vague, general overview without actionable insights
- Formatting requirements (bullet points) significantly improved readability
- Basic prompt ensured coverage of causes, effects, and solutions
- Average Score Difference: +2.4 points in favor of basic prompt
- Demonstrates critical importance of structure for summarization tasks

Scenario 4: Advice and Recommendations – Results & Analysis

Naïve Prompt: "How do I improve my productivity?"

Expected Output: General suggestions

Basic Prompt: "Give me 5 specific, actionable productivity tips for remote workers who struggle with focus. For each tip, provide a brief explanation and one practical implementation example."

Expected Output: Targeted, detailed recommendations with examples

This scenario evaluated the AI's ability to provide advice and recommendations, comparing a broad, open-ended question with a highly structured prompt that specified audience, quantity, specificity, and examples. The objective was to assess how prompt detail influences the practicality and applicability of the generated advice.

Criteria	Naïve Score	Basic Score	Difference	Notes
Clarity	3/5	5/5	+2	Basic format with examples is much clearer
Accuracy	4/5	5/5	+1	Both accurate; basic more targeted to remote workers
Completeness	2/5	5/5	+3	Naïve lacks examples; basic includes all details
Depth	2/5	5/5	+3	Basic provides implementation examples
Relevance	2/5	5/5	+3	Basic specifically addresses remote worker challenges
Structure	3/5	5/5	+2	Basic uses numbered list with explanations
Actionability	2/5	5/5	+3	Basic examples make tips immediately implementable

Key Findings:

- Basic prompt generated highly actionable, context-specific advice
- Naïve prompt provided generic suggestions without practical application
- Inclusion of implementation examples dramatically increased usefulness
- Basic prompt ensured focus on remote worker challenges
- Average Score Difference: +2.4 points in favor of basic prompt
- Shows importance of audience specification and practical examples

Scenario 5: Technical Explanation - Results & Analysis

Naïve Prompt: "Explain machine learning"

Expected Output: A basic overview

Basic Prompt: "Explain machine learning in 250 words for someone with basic programming knowledge. Include: definition, three main types (supervised, unsupervised, reinforcement), real-world applications, and key differences from traditional programming."

Expected Output: A comprehensive technical explanation with specific components

Criteria	Naïve Score	Basic Score	Difference	Notes
Clarity	3/5	5/5	+2	Basic explanation is more structured and accessible
Accuracy	4/5	5/5	+1	Both accurate; basic more technically precise
Completeness	2/5	5/5	+3	Naïve misses types and applications; basic covers all
Depth	2/5	5/5	+3	Basic explains mechanisms and real-world use
Relevance	2/5	5/5	+3	Basic addresses all specified technical aspects
Structure	2/5	5/5	+3	Basic uses clear organization for complex topic
Actionability	3/5	4/5	+1	Basic better for understanding and application

Key Findings:

- Basic prompt produced a comprehensive technical explanation
- Naïve prompt resulted in superficial overview lacking technical depth
- Specifying audience level (basic programming knowledge) improved clarity
- Basic prompt ensured coverage of all three ML types and applications
- Average Score Difference: +2.3 points in favor of basic prompt
- Demonstrates critical importance of structure for technical topics

Comparative Summary: Naïve vs. Basic Prompts

Overall Performance Comparison:

Average Scores Across All Scenarios:

Evaluation Criteria	Naïve Avg	Basic Avg	Difference
Clarity	3.2/5	5.0/5	+1.8
Accuracy	3.8/5	4.8/5	+1.0
Completeness	2.0/5	5.0/5	+3.0
Depth	2.0/5	4.6/5	+2.6
Relevance	2.4/5	5.0/5	+2.6
Structure	2.4/5	5.0/5	+2.6
Actionability	2.6/5	4.4/5	+1.8

Overall Average Score Difference: +2.2 points in favor of basic prompts

Key Observations:

- 1. Consistency: Basic prompts consistently outperformed naïve prompts across ALL scenarios and ALL evaluation criteria
- 2. Largest Gaps: Completeness, Relevance, and Structure showed the most dramatic improvements (+3.0, +2.6, +2.6 respectively)
- 3. Smallest Gaps: Accuracy showed the smallest improvement (+1.0), suggesting both prompt types can be factually accurate
- 4. Scenario Performance: All five scenarios demonstrated the superiority of basic prompts, with average differences ranging from +1.9 to +2.4 points

Key Findings & Insights

01

Prompt Clarity Dramatically Impacts Output Quality

- Basic prompts with clear instructions consistently produced higher-quality responses
- The difference was most pronounced in completeness (+3.0) and relevance (+2.6)
- Implication: Vague prompts lead to vague, incomplete responses

03

Structure and Organization Are Critical

- Basic prompts that requested specific formatting (bullet points, numbered lists) received well-organized responses
- Structure improved both clarity and usability of outputs
- Implication: Request specific formatting to enhance readability

05

Accuracy Is Less Dependent on Prompt Structure

- Both naïve and basic prompts achieved reasonable accuracy (3.8 vs 4.8)
- The main difference was not in correctness but in completeness and depth
- Implication: ChatGPT is generally accurate, but structure determines comprehensiveness

02

Specificity Drives Depth and Actionability

- When prompts specified requirements (word count, format, audience), responses were significantly deeper
- Implementation examples in prompts led to more actionable outputs
- Implication: The more specific your requirements, the better the AI can meet them

04

Audience Specification Improves Relevance

- When prompts specified the target audience (e.g., "for remote workers," "for someone with basic programming knowledge"), responses were more tailored
- Implication: Always specify your audience to get more relevant content

06

Consistency Across All Task Types

- The superiority of basic prompts held true across creative, factual, summarization, advisory, and technical tasks
- No scenario favored naïve prompts
- Implication: Structured prompting is universally beneficial

Recommendations for Optimal Prompt Design

01

Be Specific and Detailed

- Include context, constraints, and requirements
- Specify word count, format, and style preferences
- Example: Instead of "Write about AI," use "Write a 300-word introduction to AI for high school students, covering definition, applications, and ethical concerns"

03

Request Specific Formatting

- Ask for bullet points, numbered lists, tables, or other structures
- Specify headings and section organization
- Example: "Provide 5 tips in a numbered list with brief explanations"

05

Specify Output Requirements

- Define length (word count, number of items)
- Specify tone (formal, casual, technical, creative)
- Request specific elements to include
- Example: "Write 200 words in a professional tone, including statistics and real-world examples"

07

Clarify the Purpose

- Explain what you'll use the content for
- Specify the desired outcome or goal
- Example: "This will be used for a training presentation, so make it engaging and visual"

02

Define Your Audience

- Specify who the content is for (e.g., beginners, experts, specific profession)
- Tailor language and complexity to audience level
- Example: "Explain blockchain for someone with no technical background"

04

Include Examples or Reference Points

- Provide examples of desired output style or tone
- Reference similar content you've seen
- Example: "Write in a conversational tone, similar to TED Talk transcripts"

06

Ask for Implementation Details

- Request practical examples or use cases
- Ask for step-by-step instructions when applicable
- Example: "For each tip, provide one concrete implementation example"

08

Use Clear Language

- Avoid ambiguous terms
- Use specific verbs (explain, summarize, compare, analyze)
- Example: Instead of "Tell me about productivity," use "List 5 evidence-based productivity strategies"

Conclusion & Summary

Executive Summary:

This experiment conclusively demonstrates that structured, detailed prompts (basic prompts) significantly outperform vague, open-ended prompts (naïve prompts) when using ChatGPT and similar AI language models.

Key Conclusions:

01	02	03
Structured Prompts Are Superior • Basic prompts outperformed naïve prompts by an average of 2.2 points across all evaluation criteria • The improvement was consistent across all five test scenarios • No scenario favored naïve prompts	Completeness and Relevance Show Largest Gains • Completeness improved by 3.0 points (from 2.0 to 5.0) • Relevance improved by 2.6 points (from 2.4 to 5.0) • Structure improved by 2.6 points (from 2.4 to 5.0) • These metrics are critical for practical utility	Specificity Matters More Than Accuracy • Both prompt types achieved reasonable accuracy • The real difference lies in completeness, depth, and actionability • Users should focus on clarity and specificity rather than worrying about factual accuracy
04	05	
Universal Applicability • The benefits of structured prompting apply across: • Creative tasks (story generation) • Factual tasks (question answering) • Summarization tasks • Advisory tasks (recommendations) • Technical tasks (explanations)	Practical Impact • Structured prompts save time by reducing the need for follow-up clarifications • They produce more usable outputs that require less editing • They ensure all required elements are included in the response	

Final Recommendation:

Always invest time in crafting clear, detailed, structured prompts. The small effort required to specify requirements, audience, format, and examples yields significantly better results. This is one of the most effective ways to maximize the value of AI language models like ChatGPT.

The experiment successfully validates that prompt engineering is a critical skill for effective AI usage, and that basic principles of clarity, specificity, and structure dramatically improve output quality.